

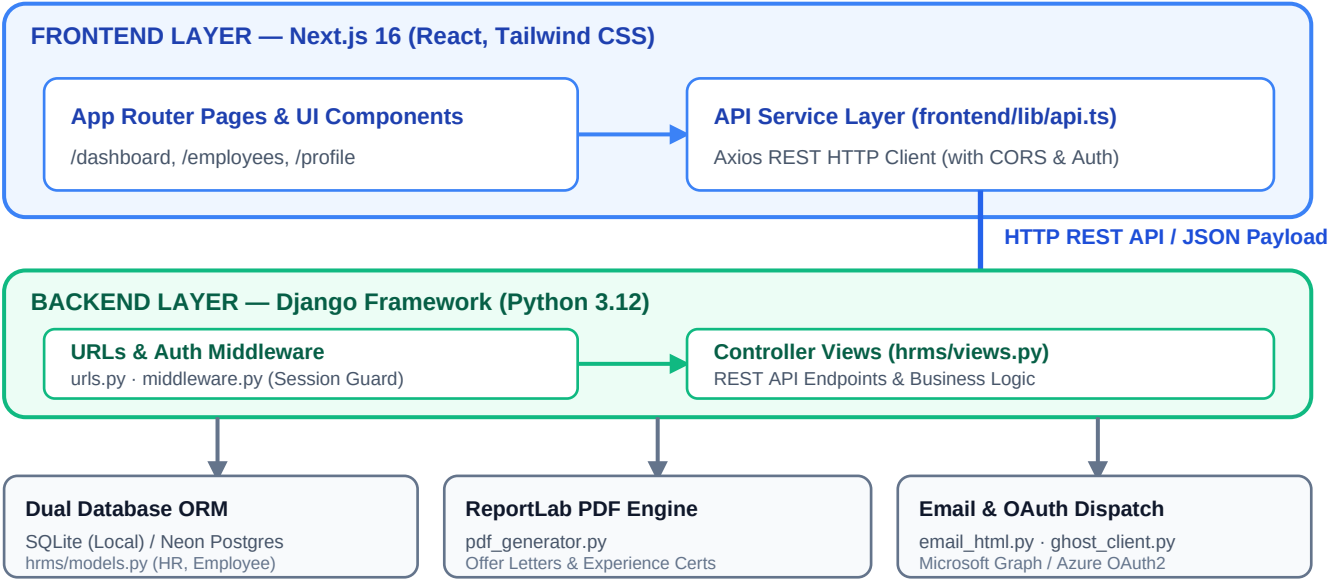
WisBees HRMS

Full-Stack Next.js & Django Architecture Reference

1. System Overview & Technology Stack

The **WisBees HRMS Application** is built on a modern decoupled full-stack architecture. The **Frontend** is powered by **Next.js 16 (React 19)** with Tailwind CSS, providing a high-performance single-page application interface. The **Backend** is driven by a **Django** REST server that manages business logic, authenticates sessions, generates automated ReportLab PDF documents (Offer & Experience letters), dispatches Microsoft Graph API emails, and connects dynamically to local SQLite or cloud PostgreSQL (Neon Postgres).

2. Frontend & Backend End-to-End Architecture Flow Diagram



3. Full-Stack Data Execution Lifecycle

- 1. Client Action (Next.js SPA):** User interacts with Next.js pages (e.g. /employees, /dashboard). React state changes trigger HTTP requests via frontend/lib/api.ts.
- 2. CORS & Auth Check:** Requests cross to Django (port 8000). corsheaders handles domain origins while hrms/middleware.py inspects session storage to attach authenticated request.user.
- 3. URL Routing & Controller Logic:** fret_project/urls.py routes requests to hrms/views.py handlers (e.g. api_employees_list, api_dashboard_stats).
- 4. ORM Querying & Dual Database:** Views perform CRUD operations against hrms/models.py. Django routes queries to local SQLite (dev) or cloud Neon PostgreSQL (prod).
- 5. PDF & Email Document Pipeline:** Offer/Experience letters are compiled into byte buffers via ReportLab in pdf_generator.py, formatted into HTML by email_html.py, and sent via Azure OAuth2 Graph API in ghost_client.py.
- 6. Payload Response:** Django responds with JSON payloads to Next.js or direct PDF binary streams for download.

4. Complete File Directory & Module Reference

File Path	Layer / Role	Description & Technical Responsibilities
frontend/app/	Next.js SPA	Next.js 16 App Router pages: /dashboard, /employees, /attendance, /announcements, /profile, /settings.
frontend/lib/api.ts	API Service	Axios REST client configured with base URL (http://127.0.0.1:8000) and credentials for session authentication.
frontend/components/	React Components	Reusable UI components: Sidebar.tsx (navigation & mobile drawer), Header.tsx (topbar title & dark mode toggle).
backend/manage.py	Django CLI	Standard Django administrative script for running dev server, migrations, and management commands.
backend/fret_project/settings.py	Backend Config	Configures dual DB routing (SQLite/PostgreSQL), CORS headers, authentication middleware, media paths, and Jinja2 template engines.
backend/fret_project/urls.py	URL Routing	Central URL directory mapping endpoints: auth (/login, /logout), REST APIs (/api/employees), and PDF generation routes.
backend/hrms/models.py	Database Models	Django ORM schemas: HR (admin auth & binary signatures), Employee (ID codes, salary, department), Attendance, Announcement, LeaveRequest.
backend/hrms/views.py	Core Controller	Primary business logic engine (70KB+) handling authentication, Next.js REST API responses, employee management, and document triggers.
backend/hrms/utils.py	Utility Helpers	Candidate ID code generators (EMP001/INT0026), password hashing, date formatters, and SMTPLib email fallback wrappers.
backend/hrms/middleware.py	Auth Middleware	Validates session cookies and binds logged-in HR Manager or Employee instance to request.user.
backend/pdf_generator.py	PDF Engine	ReportLab layout engine compiling offer letters, CTC compensation tables, terms, and digital signatures into byte streams.
backend/email_html.py	HTML Email Builder	Constructs responsive HTML email templates for offer letters, NDA packages, experience letters, and OTP notifications.
backend/ghost_client.py	Azure Email API	Handles Azure OAuth2 token generation and sends automated emails via Microsoft Graph API / Ghost CMS API.